

Under-Sampling in Digital Images

Demystifying Aliasing, Spatial Frequencies, and Signal Degradation

Image Processing & Signal Theory

What is Under-Sampling?

The Core Definition & Mathematical Threshold

Under-sampling occurs when a continuous physical image is converted into a discrete digital grid at a **spatial sampling rate lower than required** to accurately capture its fine details.

The Nyquist-Shannon Sampling Theorem

To reconstruct a continuous signal without distortion, the sampling frequency (f_s) must exceed twice the highest spatial frequency (f_{max}) present in the continuous image:

$$f_s > 2 \times f_{max}$$

- **Nyquist Rate:** The absolute minimum sampling rate required ($2 \times f_{max}$).
- **Nyquist Frequency (f_N):** The maximum bandwidth cap of a sensor ($f_s / 2$).

Key Principle: When scene details change faster across space than pixel sensors can sample, the system creates false "phantom" patterns that do not exist in reality.

Frequency Folding Mechanics

High Frequencies Disguised as Low Frequencies

When high spatial frequencies exceed the Nyquist limit (f_N), they fold back (alias) into lower frequencies of the reconstructed spectrum.

Mathematical Expression

Any frequency $f > f_N$ is erroneously reconstructed as an apparent frequency (f_{apparent}):

$$f_{\text{apparent}} = |f - k \times f_s|$$

Continuous Signal (High Spatial Frequency):

~~~~~/\~~~~~/\~~~~~/\~~~~~/\~~~~~/\~~~~~/\~~~~~/\~~~~~

Sampling Points (Too Far Apart):

\*                   \*                   \*                   \*

Reconstructed Signal (False Low Spatial Frequency):

\\_\_\_\_\_/\_

# Visual Artifacts & Degradation

How Under-Sampling Manifests in 2D Images

| Artifact                 | Primary Cause                                              | Visual Manifestation                                                       |
|--------------------------|------------------------------------------------------------|----------------------------------------------------------------------------|
| Moiré Patterns           | Interference between high-frequency textures & pixel grid. | Wavy, rainbow, or false ring patterns on fabrics, brick walls, or screens. |
| Jagged Edges ("Jaggies") | Sharp diagonal lines mapped onto coarse square pixels.     | Stair-stepped edges along curved or diagonal high-contrast borders.        |
| False Color              | Sub-sampling on single-chip Bayer color filter arrays.     | Color fringes along sharp black-and-white high-frequency boundaries.       |
| Temporal Aliasing        | Under-sampling across time (video frame rates).            | Wagon-wheel effect (wheels or rotors appearing to turn backward).          |

# 1D Spatial Signal Simulation

## Sampling Intensity Across a Line Profile

---

Demonstration of sampling a fine striped pattern at decreasing spatial sampling frequencies:

```
Position (x):  0  1  2  3  4  5  6  7  8  9 10 11 12
True Signal:  [10,  0, 10,  0, 10,  0, 10,  0, 10,  0, 10,  0, 10]  (High Freq)
-----
Adequate Sampling (fs = 1.0):
Samples:       [10,  0, 10,  0, 10,  0, 10,  0, 10,  0, 10,  0, 10]  --> Perfect Match

Under-Sampled (fs ≈ 0.67):
Samples:       [10,      0,      10,      0,      10,      0]      --> Aliased / Phase Shifted

Severely Under-Sampled (fs = 0.5):
Samples:       [10,      10,      10,      10]      --> Flat Line / Detail Lost
```

# Real-World Domain Impacts

## High-Stakes Failures Across Industries

---

- **Medical Imaging (MRI / CT Scans):** Under-sampled k-space data produces streak artifacts or geometric distortions that can mimic or obscure tumors and tissue lesions.
- **Remote Sensing & Satellite Analysis:** When pixel pitch is larger than small buildings or crop rows, sub-pixel features cause severe spectral misclassification.
- **Computer Graphics & Vision:** High-density 3D geometry rendered onto lower-resolution displays causes distracting crawling edges and shimmering textures.

# Mitigation Strategies

## Hardware Pre-Filtering vs. Software Reconstruction

---

### 1. Hardware Interventions (Pre-Sampling)

- **Optical Low-Pass Filter (OLPF):** Placed in front of the optical sensor to physically blur frequencies above  $f_N$  before light hits the pixel grid.
- **Increased Sensor Density:** Decreases pixel pitch to push the Nyquist limit past maximum scene details.

### 2. Software Interventions (Post-Processing)

- **Supersampling Anti-Aliasing (SSAA):** Renders scene at super-high resolution, applies a digital low-pass filter, and downsamples.
- **Neural Upscaling & Reconstruction:** Uses deep learning priors (e.g., DLSS) to infer true high-frequency detail from under-sampled inputs.